Full Stack Engineer — Java, Quarkus / Spring Boot & Angular | AI-Driven Microservice Architecture Design

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PROFESSIONAL SUMMARY

Full Stack Engineer with 3 years of experience at Blue Reply building mission-critical digital platforms for the consumer credit industry (Java, Quarkus, Spring Boot, Angular). My scope of responsibility has grown over time: from end-to-end feature delivery on existing platforms — calculation engines, anti-fraud checks, CMS modules — to designing greenfield microservices with modern patterns (hexagonal architecture, MicroProfile resilience, GenAI/OCR solutions), up to owning the full Quarkus microservice architecture of a multi-channel platform (37 services): surveying hundreds of APIs, C4 modeling, designing external integrations and defining team-wide standards. Hands-on across API Management (WSO2, 3scale), performance optimization and event-driven architectures (Kafka), from design through production release via CI/CD.

WORK EXPERIENCE

IT Consultant | Full Stack Engineer — Blue Reply S.r.l., Milan · 09/2023 – Present

Full stack development of web platforms and microservices for leading Italian consumer-credit and mobility groups (including Agos, Findomestic, Fiditalia and Drivalia), in Agile/Scrum teams with CI/CD on GitLab and Jenkins, and daily use of Claude Code and Cowork as AI-native development and automation tools (agentic coding, prompt engineering applied to software).

Architecture design & microservices

- Redesigned the architecture of a customer self-service platform for a leading consumer credit institution, spanning a web channel and a mobile app: surveyed 150+ REST endpoints by cross-comparing the app and web channel APIs (raw extraction, finding ~30 channel-exclusive endpoints), then remapped them onto 37 domain-driven Quarkus microservices across 3 application areas (customer data, cards, loans and financing) — ~100 endpoints exposed to the BFF across 12 business management services, 120+ external-vendor endpoints across 19 dedicated adapters, and 20 internal endpoints across 6 adapters to shared services.
- Formalized the architecture model in 6 C4 diagrams (draw.io) and 40+ sequence diagrams covering the end-to-end flows for login, customer data, loans and cards, published on Confluence with 36 components documented in linked pages: they became the shared reference across the development team, the client's architects and third-party vendors.
- Designed the integrations with 10 external systems and vendors (core banking, CRM, tracking, postal correspondence, document archiving, collections, digital-signature and OTP provider, 3D Secure ACS, message queue) applying an adapter-chain pattern with centralized authentication that isolates every external protocol (SOAP, REST, JSON-RPC) behind a uniform internal contract.
- Defined and documented the team's architecture conventions — side-by-side BFFs for web and mobile, per-domain management services, internal adapters to shared services, Kafka event processing, naming and namespaces — and industrialized documentation delivery by building a Cowork skill that automatically compiles draw.io C4 diagrams from a JSON architecture model, cutting diagram updates to minutes.
- Implemented the suite of 3 Quarkus microservices for multi-channel OTP authentication (send and validation over SMS, email and push) designed within the same model, delivering 8+ versions in 9 months: type-safe REST clients generated from OpenAPI specs, MicroProfile fault tolerance (timeouts, retries, circuit breakers), custom bean validation, health checks and a dedicated TLS truststore, with APIs exposed through WSO2 API Manager and Red Hat 3scale.

Payment Integrations — REST Exposure Layer

- Built the end-to-end Click to Pay payment flow on the REST exposure layer: 12 Camel routes and 22 Java processors for login and session management, payment initiation, case-file retrieval, barcode-event tracking and receipt download against the external payment provider, continuously shipped via GitFlow through to date.

Performance optimization — content delivery service

- Shipped 14 releases of the service (v1.1.21 → v1.1.34) in 3 months using GitFlow, reducing API gateway (WSO2) and database load by designing and implementing a version-based cache for JSON content: a virtual status endpoint exposing a file-name → version-date manifest, server-side Spring caching (synchronized @Cacheable, scheduled

eviction with hot-reconfigurable TTL), database protection rejecting requests for ids not in the catalog, and a download fallback for the web channel.

GenAI & Document OCR

- Built a GenAI web app (Angular 18 + OpenAI SDK) for LLM-assisted data extraction from identity documents, with image upload, structured JSON/Markdown output and token-usage tracking.

Loan origination platform & digital journey — full stack

- Built the IBAN blacklist anti-fraud feature end-to-end — REST services and MyBatis DAOs on the Java/Spring back end (JBoss EAP), Angular search and configuration pages on the front end — together with email/phone anti-fraud checks with control reporting and customer IBAN traceability.
- Delivered the income threshold and calculation configuration pages with the related evaluation logic, scheduled processing jobs and the remote-sale signed-contract process, over 12 months across 25+ versions (v5.1.x).
- Developed the application-routing engine of the digital journey for existing customers (outcome handling, pre-load redirection, repayment-plan remodeling) and the installment calculation services, integrating customer-registry data via OpenAPI clients and legacy SOAP, with SPID and privacy-consent management — 340+ commits in 6 months across the Angular 14 front end and a Spring Boot microservice.

Internal CMS — full stack

- Built the platform's CMS module from scratch: an Angular content-creation component (dynamic forms, partner/channel targeting, attachments) and filtered search, with Spring Boot/MyBatis APIs for search, detail and upsert; resolved 10+ UAT-reported defects.

Software Tester (Master's thesis internship) — TeamSystem S.p.A., Catania · 05/2023 – 08/2023

- Defined best practices for isolated, reproducible tests in the DevOps pipeline: database reset between runs and third-party service simulation (test doubles).
- Designed and built a containerized load-testing pipeline (Docker, Makefile) orchestrating JMeter and Prometheus/cAdvisor monitoring, with a custom Python reporting script (latency, CPU and memory charts in PDF); a single make command reproduces the full test across different load scenarios and resource limits.

TECHNICAL SKILLS

Languages: Java (8–17), TypeScript, JavaScript, SQL, Python, Dart

Backend: Quarkus, MicroProfile (Fault Tolerance, REST Client), Spring Boot 2/3, Apache Camel, microservices, REST & SOAP, OpenAPI/Swagger, OAuth2, Spring Cache, JPA/Hibernate, MyBatis, Node.js (Express), Apache Kafka

Frontend: Angular 14–18, NgRx, RxJS, HTML5, CSS3/SCSS, Bootstrap, Flutter, React (basic knowledge from personal projects)

Architecture: C4 modeling (draw.io) and sequence diagrams, solution design and API surveying, domain-driven decomposition, BFF/BFB, management, adapter and Kafka event-processing patterns, API Management (WSO2 API Manager, Red Hat 3scale), JWT

DevOps & tooling: Git, GitLab CI/CD, Jenkins, Docker, Red Hat OpenShift, Maven, Gradle, Prometheus, Jira, Confluence, Claude Code, Cowork

Testing: JUnit, JMeter, Postman, SoapUI

AI-Driven Development: Daily production use of Claude Code and Cowork (agentic coding, prompt engineering applied to software, building custom skills and automations), LLM integration (OpenAI SDK), document OCR

Methodologies: Agile/Scrum, code review, CI/CD, technical documentation

CERTIFICATIONS

Flutter Certified Application Developer — Android ATC (2024)

EDUCATION

Master's Degree in Computer Engineering (LM-32) — University of Catania · 2021 – 2023 · 110/110 with honors

Thesis: "Test analysis in the DevOps pipeline" (internship at TeamSystem S.p.A.)

Bachelor's Degree in Computer Engineering (L-8) — University of Catania · 2018 – 2021 · 104/110

LANGUAGES

Italian (native) · English (B2)

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